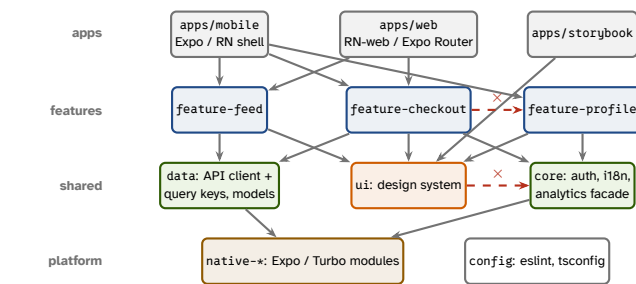
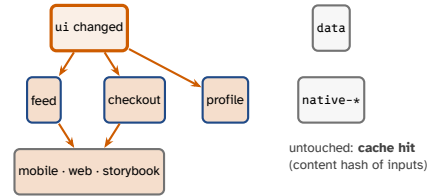


In one line: A big RN codebase stays fast to change when it is a **monorepo of packages** with **one dependency direction** — apps → features → data / UI → native modules — each package exposing a **public API** ("exports"), boundaries **enforced by lint**, an **owner** per package, a **task graph** that builds and tests only what changed, and exactly **one copy of react-native**.



grey = allowed imports, always downward; red dashed (feature→feature, ui→core) fails CI

PR touches ui → what runs?



`nx affected -t lint test typecheck -Turborepo --filter`
= the reverse dependency graph of the diff; remote cache shared by CI + laptops

The repo

- Workspaces** (Yarn, pnpm, npm, Bun) symlink packages/* into node_modules; **Nx** or **Turborepo** add a **task graph** from those dependencies, input hashing, local + **remote cache**, and "affected" runs.
- One react-native, one react:** Expo — "duplicate React Native versions in a single monorepo are not supported". Pin them in the app, peerDependencies in packages; check npm why react-native / pnpm why.
- Metro:** Expo SDK 52+ configures a monorepo automatically; bare RN adds the root to watchFolders. RN 0.79 (Metro 0.82) resolves package.json "exports" and "imports" by default (opt out: resolver.unstable_enablePackageExports).
- pnpm:** isolated node_modules catches *phantom* (undeclared) deps; Expo supports it from SDK 54, else nodeLinker: hoisted.
- Native code autolinks from the app:** a native dependency must be resolvable from apps/mobile (Expo SDK 54 autolinkingModuleResolution, automatic in SDK 55).

Packages and boundaries

- Feature package** = vertical slice: screens, its navigator, hooks, query keys, analytics events. Public surface = index.ts + "exports"; deep imports (feature-feed/src/x) are not resolvable.
- Features never import features.** Cross-feature needs go *down* (a shared data/core package), *through navigation* (route + params), or through an injected interface the app shell wires up.
- Enforce:** Nx tags (type:feature, type:ui, scope:checkout) + @nx/enforce-module-boundaries depConstraints; or no-restricted-imports, import/no-cycle; TS project references for build order.
- Ownership:** CODEOWNERS per package path → the owning team must approve; a package without an owner is where rot starts.
- State by owner:** server cache (TanStack Query) keys *owned* by the data package — invalidation from any feature uses the same key factory; client stores (Zustand) *per feature*; truly global = session, theme, flags only.

Example — public API, boundary rule, thin route

```
// packages/feature-feed/package.json
{ "name": "@acme/feature-feed",
  "exports": { ".": "./src/index.ts" }, // nothing else
  "peerDependencies": { "react": "*", "react-native": "*" } }
// eslint.config.mjs (root)
@nx/enforce-module-boundaries: [ 'error', { depConstraints: [
  { sourceTag: 'type:feature',
    onlyDependOnLibsWithTags: [ 'type:data', 'type:ui', 'type:core' ] },
  { sourceTag: 'type:ui', onlyDependOnLibsWithTags: [ 'type:ui' ] } ] ] }
// apps/mobile/app/(tabs)/feed.tsx -- Expo Router route file
export { FeedScreen as default } from '@acme/feature-feed';
```

Versioning, build, release

- Internal packages: "workspace:*" — no versions, always head; only packages *published* outside (an SDK, the design system for another repo) get semver + Changelists.
- tsc -b with project references (or Nx typecheck) per package — incremental, and a cycle is a build error.
- One native build per app; OTA channel per app + runtimeVersion; a feature flag per risky feature so a bad feature is switched off, not hot-fixed.
- Polyrepo** instead? Only when teams must release independently — the price is version skew, cross-repo PRs and a published package for every shared change.
- Migrating a big app in:** start with config + ui, then carve one feature at a time out of apps/mobile/src (strangler) — boundaries as *warnings* first, errors once the count reaches zero.

Navigation architecture

- The **app shell owns the root** (auth gate, tabs, modals); each feature exports its screens or a nested stack + typed ParamList; the shell composes them and merges the features' deep-link config.
- Expo Router:** routes are files in the app's app/; keep them one-line re-exports so features stay router-agnostic and testable.
- Cross-feature jumps by **route name + params** (or a URL), never by importing a screen component.

Universal (iOS · Android · web)

- react-native-web** renders View/Text/Pressable as DOM; **Expo Router** gives Android, iOS and web one file-based navigation, with **static rendering** on web (SEO).
- Platform files: Map.native.tsx / Map.web.tsx (also .ios / .android) — the bundler picks; the *interface* lives in the shared file's types.
- Share logic and UI primitives; accept per-platform screens where UX differs (hover, keyboard, URL bar). Web bundle size and SSR-unsafe code (window) are the tax.

Native modules package

packages/native-*: Expo Modules API (Swift/Kotlin) or a TurboModule spec + Codegen; a typed JS facade, a **web / Jest fallback**, its own changelog. Native changes bump the app's runtimeVersion — OTA updates can't ship them.

Brownfield — RN inside an existing native app

- Official path: native project moves into ios/ + android/ of an RN project; host screens via RCTReactNativeFactory (iOS) / ReactActivity + ReactHost (Android). One shared runtime; each RN screen is an AppRegistry name.
- At scale: ship RN as a **prebuilt XCFramework / AAR** (e.g. Callstack's react-native-brownfield CLI) so native teams need no Node toolchain; native keeps the navigation; data via initial props + a native-module/event contract.

Traps

- Two react copies → "Invalid hook call"; two react-native → native module / view registered twice. Dedupe, peer-depend.
- A shared/common package everything depends on — a hidden monolith; split by purpose (ui, data, core).
- Barrel index.ts re-exporting a heavy module eagerly — startup cost in every importer.
- Native dep declared only in a feature package — not autolinked, crash at runtime.
- Hoisting hides undeclared deps until a CI or pnpm install breaks them.

Remember

Apps → features → shared → native, never sideways · exports = the API · lint enforces · CODEOWNERS own · affected + cache · one react-native.